ISSN: 1992-8645 www.jatit.org E-ISSN: 1817-3195

EMOJI_SENTI_WORD: AN ENHANCED METHOD FOR EMOJI SENTIMENT CLASSIFICATION AND POLARITY SHIFT HANDLING USING EMOJIS

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ABSTRACT

The increased use of emojis in online textual communication highlights the critical need for their effective classification in sentiment analysis to extract valuable information. Since most social media posts are very short, a careful understanding of emoji usage is necessary to effectively handle the polarity shift/ change in the overall sentiment. Additionally, the voluminous social media data requires a more efficient sentiment classification system with greater accuracy and speed for modern real-time applications. Therefore, this research aims to analyse the impact of emojis on sentiment classification, especially when they act as catalysts for polarity shifts in sentences, and develop an enhanced method for emoji classification that effectively handles polarity shifts with better accuracy and speed compared to existing systems/ methods.

Nine Machine Learning classifiers and the Long Short-Term Memory (LSTM) Deep Learning model were trained on a dataset of corona-related tweets. The inclusion of emojis in the text resulted in a notable increase in accuracy from 67% (text only) to 84%. For tweets where emojis caused polarity shifts, accuracy increased to 97%, even when emojis alone were used, and the processing time was reduced by half as well. Among the classifiers, LSTM demonstrated high accuracy.

To further enhance emoji sentiment classification, a new method named ‘EMOJI_SENTI_WORD’ was proposed in this research, which improved accuracy up to 10% across various classifiers. In contrast to the traditional method, ‘EMOJI_SENTI_WORD’ replaced emojis with sentiment words based on their sentiment polarity intensity. This unique approach enables EMOJI_SENTI_WORD to achieve improved accuracy and performance. Apart from the corona-related dataset, the EMOJI_SENTI_WORD algorithm was also tested on two additional tweet datasets: ‘MovieReview’ and ‘OnlineClass’, to validate its scalability and adaptability, achieving a notable 8% increase in accuracy.

Keywords: *Sentiment Analysis, Emojis, Emoticons, Polarity Shift, Opinion Mining*

1. INTRODUCTION

Social media serves as a powerful communication platform that has greatly diminished emotional disconnection among people, even during the era of global COVID-19 social distancing measures. The widespread growth and extension of social media usage worldwide has transformed the way people communicate with each other and share their opinions, feelings, emotions, and thoughts. The number of individuals engaging with social media continues to grow annually. According to statistics published in February 2025 [1], 63.9% of the global population uses social media, with mobile phones being the dominant device for internet access. This accessibility makes it an attractive ‘anytime’, ‘anywhere’ communication platform for fast and effortless information exchange. Twitter, one of the prominent social media platforms (which is now rebranded as X), has more than 245 million daily active users, as reported in Twitter statistics by Demand Sage [2]. The huge amount of data generated by these users inspires researchers, motivating them to analyse this data to extract meaningful information that may fuel the development of intelligent real-world applications.

Journal of Theoretical and Applied Information Technology
31st October 2025. Vol.103. No.20

ISSN: 1992-8645 www.jatit.org E-ISSN: 1817-3195

In this realm of research, 'Sentiment Analysis' or 'Opinion Mining' [3, 4] plays a vital role, facilitating the understanding of people's opinions, feelings, emotions, and attitudes towards diverse topics and entities.

1.1 Usage of Emojis and Emoticons

In today's context, the majority of online messages exchanged through social media widely use both emojis and emoticons because of their ability to compensate for non-verbal cues, which are typically absent in online textual communication [5]. These powerful communication tools have a huge influence on the effectiveness of the opinion mining process. For this research work, tweets incorporating emojis and/or emoticons were collected.

An emoticon is an abbreviation for 'EMOTion ICON', and it refers to a pictorial representation formed using keyboard characters like punctuation marks, numbers, and letters, for expressing a person's feelings, moods, and emotions [6]. Some examples of traditional emoticons include :-), :-], :-} for a 'happy face', :-)) for a 'very happy face', :'-(for a 'crying face', :-(, :-[, :{ for depicting emotions such as sadness, frowning, anger, and so on.

Emojis, often considered an advanced version of emoticons, are graphical images used to express one's feelings and thoughts. Emojis are available in different genres like facial expressions, animals, fruits, flags, places, symbols, and everyday objects [7]. According to 2022 statistics [8], the emojis 😂 (Face with tears of joy) and 😭 (Loudly crying face) were the most frequently tweeted. Some other popular emojis on Twitter across 2020 and 2021 included [8, 9]: 🥺 (Pleading face), 🤣 (Rolling on the floor laughing), 😍 (Smiling face with heart_eyes), 😔 (Pensive face), 🥰 (Smiling face with hearts), 🤔 (Thinking face), 🙄 (Face with rolling eyes), and 😊 (Smiling face with smiling eyes). The presentation style of emojis varies among different operating systems and social network platforms, potentially influencing user preferences for emojis on these platforms [10]. These emoji characters are encoded in Unicode (Universal Coded Character Set), which is a standard in the computing industry for consistent character encoding [11].

1.2 Effects of Emojis on Polarity Shifting

Polarity shift is a challenge in sentiment classification problems, where the sentiment polarity of a text is reversed. This phenomenon greatly affects the accuracy of sentiment classification systems. As shown in Table 1, the presence of emojis in a tweet also causes such polarity shifts [12], and this is the main focus of this research. Sarcastic texts can also be identified, where the pure text sentiment contradicts the sentiment of the text along with emojis.

Table 1: Presence of Emojis Causing Polarity Shift

Sample tweet	Polarity of text without emoji	Polarity of text with emoji (causing polarity shift)
You can't even cough around people no more without them thinking you got corona😂💯 😂🤣😂 I'll die laughing before I die of corona	Negative	Positive (Sarcasm)
I want to see it 😭 This is how corona has got me feeling everyday😭	Positive	Negative
After Corona Pandemic, The Revival Of Tamil Cinema Industry Was Done By This Man 🖤! Just booked my corona vaccine appointment 🥰🥰🥰🥰🥰	Neutral	Positive
And corona just started a year ago😭 Corona Virus Cases in Sweden 📅 Date = 2021-04-14 😕 Confirmed Cases = 885385 😥 Deaths = 13720	Neutral	Negative

Due to the prevalent use of emojis and emoticons in contemporary online communication and their influence on polarity shifts in text, there is a pressing need to evolve an enhanced method for emoji sentiment classification to achieve higher levels of accuracy.

The main contributions of this research work are presented below:

- Automatic text classification was performed using the VADER (Valence Aware Dictionary and sEntiment Reasoning) sentiment analysis tool, and its effectiveness was also assessed by comparing with manual classification.
- The effects of emojis and emoticons on sentiment classification were analysed.
- The effects of emojis on polarity shifting were also demonstrated. For such a subset of tweets, a method to increase accuracy while reducing execution time was also discussed. Additionally, the execution time of this method was compared to that of the traditional method.
- A novel method, named 'EMOJI_SENTI_WORD', was proposed for effective emoji sentiment classification.

The outline of various sections of this paper is given as follows: After reviewing prior research on sentiment analysis with emoticons and emojis in Section 2, a detailed explanation of the various steps involved in the research process is given in Section 3. The theoretical and visual presentation of the research outcome is discussed in Section 4, and the last Section (Section 5) concludes with a summary of the important findings and provides recommendations for future research.

2. RELATED WORKS

This section of the paper presents various research works on sentiment classification with emojis and emoticons, using both supervised and unsupervised Machine Learning (ML) methods.

2.1 Sentiment Classification with Emoticons

In the early days, research works focussed only on the role of emoticons in sentiment classification. A select few of these research works are discussed in this sub-section.

Min *et al.* [13] demonstrated the usage of emoticons in sentence-level sentiment classification and its impact on Precision and Recall for sentiment extraction. They identified the frequency of each emoticon, revealing that despite the huge number of emoticons in social media, only a small subset was used frequently. Most of these frequently used emoticons were observed to signal positive emotions. Two corpora were used in this experiment: SelectCorpus, a human-annotated corpus consisting of sentiment-rich sentences selected from their sentiment analysis system's output, and RandomCorpus, comprising randomly collected sentences that were not sentiment-rich. In the case of SelectCorpus, a precision rate of 75.2% was achieved, indicating that emoticons could be considered a good indicator of sentiment. The precision rate dropped to 40.1% in the case of RandomCorpus, suggesting that emoticons alone might not be trusted as a sentiment indicator, but they could be used for sentiment classification, especially when other linguistic clues were lacking.

Emoticon Space Model (ESM), a semi-supervised model proposed by Jiang *et al.* [14], was used to identify the subjectivity, polarity, and emotion of microblog posts. An emotion space was first constructed using the 100 most frequently used emoticons, which was followed by the Projection and Classification phases. In the Projection phase, posts were projected into the emotion space by considering the semantic similarity between the emoticons and the words. In the Classification phase, the coordinates of the posts were used for supervised sentiment classification tasks. The dataset used in this research was a public Chinese microblog benchmark corpus. Among the baseline models, Support Vector Machine (SVM) demonstrated the best performance in both polarity and subjectivity classifications. However, ESM outperformed SVM in all three classification categories: subjectivity, polarity, and emotion.

Another work on multilingual data [15] focussed on sentiment analysis of Greek documents. A semi-supervised technique was followed, which considered both emoticons and a list of emotionally intense keywords separately to evaluate performance. The method of using keywords achieved 90% accuracy in identifying the subjectivity level and 93% accuracy for the polarity level. On the other hand, the technique of using emoticons achieved accuracies of 74% and 77% for subjectivity and polarity levels, respectively.

The analysis by Wang and Castanon [16] demonstrated the wide usage of emoticons, especially some of the positive sentiment emoticons used by Twitter users. The relationship between emoticon usage and sentiment polarity was analysed in different ways. In a survey conducted to know the perceived sentiment polarity of the most frequently used emoticons, it was identified that certain emoticons strongly signalled sentiment polarity while others were ambiguous. Clustering of emoticons and words expressing similar sentiments led to the discovery that the meaning of some complex emoticons could be better understood by considering the words that appeared in the same context. Comparing the perceived sentiments of tweets before and after removing the emoticons revealed that in almost half of the tweets, emoticons were the only component signalling sentiment strongly, and their removal made the classifiers inaccurate.

In the research by Bahri *et al.* [17], a novel algorithm called 'emoticon score learning' was proposed to find out the score of emoticons based on the context of the tweets in which they were used. The sentiment classification of tweets was performed by taking into account both the text-score and emoticon-score of tweets. If both were positive, the tweet was considered a positive tweet. If both were negative, the tweet was negative. If the scores contradicted each other, the tweet was considered a 'both-sentiments' tweet. The seven most frequently used emoticons alone were considered in this research work. The proposed algorithm was evaluated on 1000 tweets extracted from the internet and produced an accuracy of 91.1%.

Ullah *et al.* [18] researched airline data extracted from Twitter using machine learning techniques, such as Support Vector Machine (SVM), Random Forest (RF), Naïve Bayes (NB), and logistic regression, and deep learning techniques, such as Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN). The proposed algorithm considered both emoticons and text for sentiment analysis. In the first step, sentiment was identified with emoticons in the text, and in the second step, sentiment was identified after removing emoticons and punctuation from the text. Around 1-3% of accuracy difference alone was observed when both methods were compared. The performance of deep learning models was better than that of machine learning models when both emoticons and text were considered.

A method for polarity shift detection has been introduced in the research by Ayeste and Noferesti [19], in which a corpus with polarity shift-tagged sentences was constructed automatically using the concept of distant supervision. The polarity of emoticons, hashtags, and sentence words was used to identify polarity shifts in sentences. Based on these semantic features, the SVM machine learning classifier was trained on the constructed corpus and found to detect polarity shifts with an accuracy of 79.33%.

In the research mentioned above, only a limited number of emoticons were considered. Many studies focussed on comparing the efficiency of various algorithms, and there seemed to be a minimal (or no) improvement in performance when emoticons were used along with the text. An overview of research works involving emoticons alone has been provided in this section, and the following sub-section will delve into various research related to sentiment analysis using emojis.

2.2 Sentiment Classification with Emojis

The research by Hankamer and Liedtka [20] demonstrated the positive impact of handling emojis on the performance of sentiment analysis tasks. Tweets containing at least one emoji from the 400 most popular emojis were collected from Twitter and were constructed into an Emoji Dataset. This dataset was labelled using VADER sentiment analyzer and trained on an existing and popular Twitter sentiment analysis dataset - 'Sentiment140'. This training was performed on the original 'Sentiment140' dataset and then on the Emoji Dataset with and without emojis. Naïve Bayes, Maximum Entropy, Support Vector Machine, Shallow Neural classifier, and Recurrent Neural Network models were used for evaluation, and there were performance gains across all models when emojis were included.

Another study conducted by Shiha and Ayvaz [21] also examined the effects of emojis on sentiment analysis and text mining, and they found that the usage of emojis improved the sentiment score. Data from Twitter related to two important events were collected: 'New Year Eve' (a positive event) and 'Istanbul attack' (a negative event). The sentiment score was calculated using the words present in the 'SentiWordNet' dictionary in two different ways: first, by removing emojis from the tweets, and second, by including emojis in the tweets, for both positive and negative events. As a result of this analysis, the presence of emojis was found to have a greater influence on the overall sentiment score of positive opinions when compared to negative opinions.

The impact of emojis on the accuracy of emotion classifiers was studied in another work [22]. Initially, tweets were classified into seven emotion classes (sad, angry, happy, scared, thankful, surprised and love) based on the automatic labelling of hashtags. When the input tweets were trained with and without emojis on Multinomial Naïve Bayes (MNB) and SVM classifiers, there was a slight increase in accuracy when emojis were considered. It was also suggested to use the Bag of Words representation with the MNB classifier for large vocabularies to achieve better sentiment classification accuracy.

In the research by Karthik *et al.* [23], various Machine Learning classifiers, along with the 10-fold cross validation technique, Artificial Neural Networks (ANN), and Convolutional Neural Networks (CNN), were trained on the tweets extracted from Twitter to predict the polarities of emojis. The standard text mining classifier, TextBlob, was used for mining text sentiment. The polarities of both the text and emojis were then aggregated by taking their average to find the final polarity of the tweet. The difference in polarities between the text and emojis was used to detect sarcastic statements in some of the tweets. The machine learning classifiers gave very poor performance with accuracy ranging from 6.83% (for Naïve Bayes) to 36.33% (for Support Vector Machine) on polarizing emojis, whereas the accuracies of ANN and CNN were found to be 48.95% and 97.43% respectively.

The work by Li *et al.* [24] considered the popularity of emojis on Twitter and analysed the feasibility of an emoji training heuristic for multi-class sentiment classification. This heuristic automatically prepared the training dataset and classified the tweets into five classes: Highly Positive, Positive, Neutral, Negative, and Highly Negative, by adding up the sentiment score of all the emojis in the tweets. To evaluate the performance, several machine learning classifiers were employed on the test data after pre-processing and feature extraction. The use of logistic regression resulted in better performance.

To avoid the human effort in (manually) annotating the data, a few other research works concentrated on unsupervised methods. Kralj Novak *et al.* [25] claimed to be the first provider of an emoji sentiment lexicon called 'Emoji Sentiment Ranking', which was the first publicly available resource. Over 1.6 million tweets extracted in 13 different European languages were labelled by 83 human annotators, and out of these, only 4% had emojis. In the process of creating a lexicon using Emoji Sentiment Ranking, 751 of the most frequently used emojis were given sentiment scores by considering the sentiment of the tweets in which they were present. Some interesting findings emerged from this research: most of the popular emojis were positive, emojis were identified to be present at the end of tweets, and the sentiment intensity also increased with distance.

Another research work [26] evolved a simple approach to automatically construct an emoji sentiment lexicon with the sentiment words extracted from the available English dictionary 'WordNet-Affect'. The co-occurrence frequency between these extracted sentiment words and the emojis present in the input tweets was calculated. Based on the frequency of occurrence of emojis, they were assigned a multi-dimensional vector whose elements depicted the sentiment strength. The effectiveness of this approach was demonstrated by comparing this lexicon with the conventional lexicon for both three (positive, negative and neutral) and five (happiness, disgust, sadness, anger and fear) sentiment categories.

A similar work [27] constructed an emoji sentiment lexicon automatically with 840 emojis based on the definitions given in Emojipedia by its creators, using an Unsupervised System with Sentiment Propagation Across Dependencies (USSPAD) approach. This approach was further improved by different variants that considered the sentiment distribution of informal texts along with emojis. Applying these approaches to both Spanish and English datasets led to performance improvements when compared to annotated datasets used in some other research studies.

In another research work on polarity shift handling [28], it has been demonstrated that the effective handling of complex polarity shift patterns, such as negations, intensifiers, diminishers, and contrast transitions, results in improved accuracy in sentiment classification tasks.

Based on the study of various research works on sentiment analysis as given above, it is observed that most of them concentrated either on emotions or emojis. Also, some of them compromised on speed for better accuracy in their proposed approaches. Some approaches lacked proper handling of polarity shifts due to restricted emoji/ emoticon classification. This reveals the gap in current approaches regarding scalability, performance, accuracy, and adaptability for modern real-world applications.

Therefore, the motivation for this proposed research is to analyse the significance of effectively handling emojis and emoticons when dealing with polarity shift problems, ultimately to enhance sentiment classification and also to evolve a new method that further improves accuracy. While the research mentioned above calculated the sentiment score of emojis based only on the

ISSN: 1992-8645 www.jatit.org E-ISSN: 1817-3195

sentiment of the text or tweet, the new method proposed in this research aims to replace emojis with appropriate sentiment words that clearly describe their meaning for better emoji sentiment classification.

The focus of this research work is on supervised machine learning classifiers since unsupervised methods need substantial effort and time to build lexicons, patterns, and rules [29]. However, measures have been taken to enhance the performance of the sentiment classification system by training it on the most commonly used emojis and emoticons. The dataset used in this research work contains around 11,200 instances of emojis, comprising 521 unique emojis. Importantly, this dataset also includes all the popular emojis from recent years on Twitter, as mentioned in Section 1.1.

3. METHODOLOGY

This section provides a detailed explanation of the research procedure undertaken to address the following objectives:

- To analyse the impact of emojis on sentiment classification
- To demonstrate the influence of emojis on polarity shifting
- To propose a new method ('EMOJI_SENTI_WORD'), aimed at enhancing the emoji sentiment classification process

As the objectives are accomplished, the efficiency of the traditional method for emoji classification is compared with the proposed method as well. The block diagram in Figure 1 shows the outline of the steps involved in this research work.

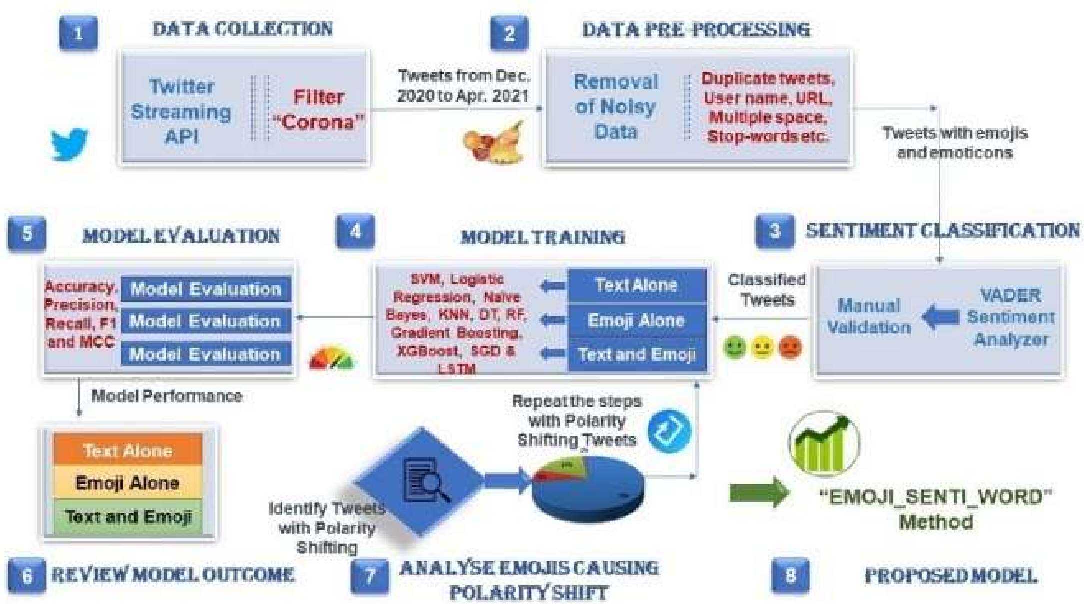


Figure 1: Block Diagram Showing the Research Steps

3.1 Data Collection

The tweets were collected from Twitter, employing the filter 'corona'. Out of a total of 90,139 collected tweets, this research work considered 5,000 tweets with emojis and emoticons.

3.2 Data Pre-processing

Pre-processing involves removing undesirable, unnecessary, and unpredictable noisy data from raw, unstructured input datasets, making the data more meaningful for further analysis and inference. This process becomes more important when dealing with data taken from social networks [30]. The following actions were performed at the time of pre-processing:

- Unnecessary columns from the dataset, duplicate tweets, User-Name tags, RT-Tags, URLs, all single characters, and stop-words like 'it', 'he', 'she', 'is', and 'the' were removed.
- Single space was substituted for multiple spaces.
- Lemmatization was done, where various forms

Journal of Theoretical and Applied Information Technology
31st October 2025. Vol.103. No.20

ISSN: 1992-8645 www.jatit.org E-ISSN: 1817-3195

of a word were converted into their root or dictionary form to get better results. For example, words like ‘contains’, ‘contained’, and ‘containing’ were converted into the root form ‘contain’.

After data pre-processing, the total number of tweets was reduced from 90,139 to 39,555, wherein 5,000 tweets containing emojis and emoticons alone were considered in this research work.

3.3 Sentiment Classification

Labelling all the collected tweets manually would be a time-consuming process. Therefore, VADER (Valence Aware Dictionary and sEntiment Reasoning) sentiment analyzer was used to find the sentiments of the tweets. VADER is a rule-based sentiment analysis model that has been proven to be effective in classifying social media texts [31].

The score VADER produces indicates sentiment polarity (positive/ neutral/ negative) as well as sentiment intensity, measured on a scale from -1 to +1, where -1 means extreme negative and +1 means extreme positive. Subsequently, sentiment was calculated from the compound score, as given in Table 2.

Table 2: Sentiment Calculation from Compound Score

Compound Score	Sentiment
>= 0.05	Positive
> -0.05 and < 0.05	Neutral
<= -0.05	Negative

The following features in VADER sentiment analyzer have made it an effective model for classifying social media text.

- *Punctuation* marks like exclamation (!) increase the intensity of sentiment in a sentence. For example, the sentiment intensity for the sentence ’The ice cream is tasty!!’ is higher than the intensity of the same sentence ‘The ice cream is tasty’ without the exclamation.
- *Intensifiers and Diminishers* (called Degree Modifiers) also influence the intensity of sentiment. For instance, in the sentence ‘The ice cream of this brand is very tasty’, the intensifier ‘very’ increases sentiment intensity, while ‘rarely’ in ‘The ice cream of this brand is rarely tasty’ reduces it.
- When a sentiment word is fully *capitalised* while other words are not, sentiment intensity increases. For example, the sentiment intensity is higher for ‘The ice cream is TASTY’, compared to ‘The ice cream is tasty’.
- *Conjunctions* like ‘but’ that change the polarity of a sentence are also handled effectively in VADER.
- VADER has also been designed to handle emojis, emoticons, abbreviations, and slang in a sentence.

Figure 2 presents the distribution of tweets in the dataset after sentiment classification, across three classes: Positive, Negative, and Neutral.

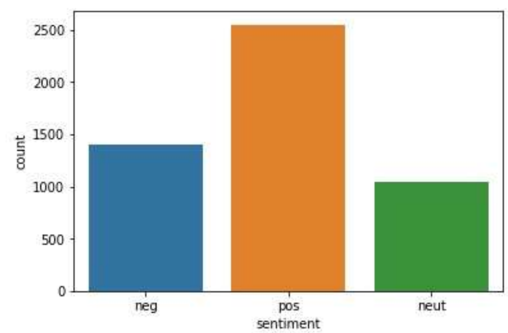


Figure 2: Classification of Tweets into Three Sentiment Classes

To assess the accuracy of the automated sentiment classification done by the VADER sentiment analyzer, 30% of the input tweets were manually labelled and then compared with the automated labels generated by VADER. This comparison showed that 82% of the tweets had the same labelling between the manual and automated classifications.

3.4 Training ML Models

After the automated classification of tweets, the next step was to train the labelled dataset using several machine learning models. Since one of the main objectives of this research work was to analyse the impact of emojis in detecting and handling polarity shift problems, the training was carried out using the following three sets of the original tweet dataset:

- Text_alone: In this first set, pure text alone was extracted from the tweets, and the ML models were trained.
- Emoji_alone: Emojis and Emoticons alone were extracted from the tweets, forming a separate set for training.
- Text+Emoji: The original dataset of tweets with both text and emojis was used for training the ML models.

A sample tweet represented in each of the above

ISSN: 1992-8645 www.jatit.org E-ISSN: 1817-3195

three sets is given below:

Text_alone	Emoji_alone	Text+Emoji
Tier four lockdown be safe guys	😬 😷	Tier four #lockdown 😬 be safe guys 😷

Vectorization

The TF-IDF ('Term Frequency – Inverse Document Frequency') vectorizer was used in this research to transform textual data into a numerical representation [32]. This vectorizer measures the importance and relevance of a word in a document by multiplying the following two measures:

$$TF = \frac{Number\ of\ times\ the\ term\ or\ the\ word\ appears\ in\ a\ document}{Total\ number\ of\ terms\ in\ the\ document}$$

and

$$IDF = log_{10} \frac{Total\ number\ of\ documents}{Number\ of\ documents\ in\ which\ the\ term\ appears}$$

As the score increases, the term is considered to be more relevant within the document. In the context of this research, tweets are considered as small documents.

ML optimization

One of the functions of Scikit-Learn's (sklearn) model_selection package, 'GridSearchCV', was used for hyperparameter tuning, in which different parameter values were first set before training each model [33]. Unlike the random search approach of parameter tuning ('RandomizedSearchCV'), the grid search method exhaustively tries every possible combination of parameter values, aiming to find the combination of hyperparameter values that results in maximum accuracy. This 'GridSearchCV' was used with the k-fold cross validation technique, with a cross validation (cv) value of 5. In this process, the dataset was divided into k non-overlapping folds or sets, where k-1 sets were used for training, and the remaining one was used for testing. The model was then tested and trained k times, each time with a distinct set for testing and training. The final cross-validated score was determined by taking the average of the results obtained across all k iterations.

ML models

The following Machine Learning models [34-37] were used to evaluate performance on the three different sets (Text_alone, Emoji_alone and Text+Emoji) of the original dataset.

- Support Vector Machine (SVM): In SVM, classification is done by finding an optimal hyperplane within an n-dimensional space (where n is the number of features), to separate the data into various target classes. The hyperparameters tuned are the regularization parameter ('C'), kernel type to be used, namely linear, radial basis function, polynomial, sigmoid ('kernel'), and kernel coefficient for handling non-linear classification ('gamma').
- Logistic Regression: This algorithm predicts a categorical dependent variable using one or more independent variables. The sigmoid or logistic function is used for mapping predicted values to probabilities. In binomial or binary logistic regression, the dependent variable will have only two possible outcomes, whereas in multinomial logistic regression, the target or dependent variable can have three or more possible outcomes. The hyperparameters tuned are 'penalty' to specify the norm for penalization, and 'C', which is the inverse of the regularization strength.
- Naïve Bayes: This classifier is based on Bayes' theorem, which works on the principle of conditional probability. It is called 'naïve' because the presence of a certain feature in a class does not relate to the presence of any other feature. The additive smoothing parameter 'alpha' has been tuned in this research.
- K-Nearest Neighbor (KNN): KNN assumes that the data points nearer to each other belong to the same class, and it classifies new data points based on this similarity. The parameter for setting the number of neighbors of a new data point ('n_neighbors') has been tuned with different values.
- Decision Tree (DT): It is used to predict the value of a target variable by learning simple decision rules inferred from the data features. Internal nodes of the tree denote the features to be tested, branches denote the decision rules, and the leaf nodes denote the final outcomes. The important parameter that has been tuned is the maximum depth of the tree ('max_depth').
- Random Forest (RF): Random Forest is an ensemble learning method that constructs multiple decision trees using subsets of the given dataset, gets the predictions of all decision trees, and makes a final prediction based on the majority votes of the tree predictions. The hyperparameters tuned are the number of trees in the forest ('n_estimators'), maximum depth of the tree ('max_depth'), maximum number of features to consider for the best split ('max_features'), and function to

Journal of Theoretical and Applied Information Technology
31st October 2025. Vol.103. No.20

ISSN: 1992-8645 www.jatit.org E-ISSN: 1817-3195

measure the quality of the split (‘criterion’).

- Gradient Boosting: These boosting algorithms also use the concept of ensemble learning that combines multiple simple models known as weak learners (typically decision trees) to produce a final output. It trains many models iteratively and sequentially, with each subsequent model trying to correct the errors of the previous model, resulting in performance improvement over iterations. The parameters tuned are, learning rate to control the contribution of each tree (‘learning_rate’), minimum number of samples needed to split an internal node (‘min_samples_split’), minimum number of samples needed to be at a leaf node (‘min_samples_leaf’), number of samples to be used for fitting each tree (‘subsample’), random state to control the random seed given to each tree estimator at each iteration (‘random_state’), ‘n_estimators’, ‘max_depth’, and ‘max_features’.
- Extreme Gradient Boosting (XGBoost): It is an open-source implementation of the gradient boosting algorithm. Since it supports parallel computations on decision trees, the performance and speed are generally very high compared to the gradient boosting algorithm. This uses cache optimization for resource management and distributed computing for larger datasets. Several hyperparameters tuned are ‘learning_rate’, ‘n_estimators’, ‘max_depth’, ‘subsample’, and the number of features used to fit each decision tree (‘colsample_bytree’).
- Stochastic Gradient Descent (SGD): These algorithms are a variant of gradient descent algorithms that can be used when dealing with large datasets. As only a small random subset of observations is taken at every step of the algorithm instead of the entire dataset, computation time is reduced. Some of the hyperparameters tuned are the loss function to be used (‘loss’), the regularization term used in the model (‘penalty’), and the constant that multiplies the regularization term (‘alpha’).

Deep Learning model

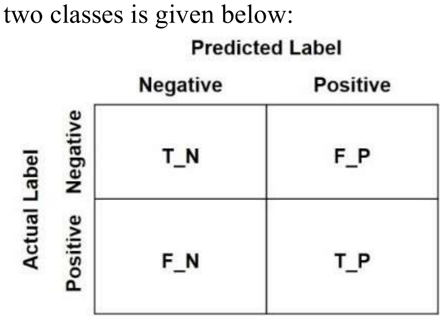
Long Short-Term Memory (LSTM), a type of Recurrent Neural Network (RNN), is widely used in contemporary sentiment analysis tasks [38]. In this research, it was also used to evaluate the performance of the original dataset on three different sets (Text_alone, Emoji_alone and Text+Emoji). LSTMs can process sequential data and are capable of learning long-term dependencies.

Some details about building the LSTM model are given below. The first layer was the embedding layer, where each word was represented by a vector of size 100. SpatialDropout1D was used to regulate the network. The next layer was the LSTM layer with 100 neurons. The output layer had 3 cells, representing the three different categories of sentiment labels (pos, neg, neut). The model was compiled using the ‘adam’ optimizer, ‘categorical_crossentropy’ as the loss function, and ‘softmax’ activation function for multi-class classification.

3.5 Model Evaluation

The following metrics were used to evaluate the performance of Machine Learning models [39, 40] on the three different sets from the input dataset.

- ‘Confusion Matrix’ presents the accuracy of a model for both binary and multi-class classification problems. This can be described as an N x N matrix, where N is the number of target classes. This matrix depicts the predicted target values of the machine learning model against the actual target values. The ‘confusion matrix’ for a binary classification problem with ‘Positive’ and ‘Negative’ as the two classes is given below:



T_N - True Negative; F_P - False Positive
F_N - False Negative; T_P - True Positive

- ‘Accuracy’ is the ratio of the number of correct predictions to the total number of predictions made.
- ‘Precision’ or ‘Positive Predictive Value’ is the fraction of relevant predictions among the retrieved predictions.
- ‘Recall’ or ‘Sensitivity’ or ‘True Positive Rate’ is the fraction of relevant predictions that are retrieved correctly.
- ‘F1-Score’ is the harmonic mean of ‘Precision’ and ‘Recall’. This measure is desirable when the class distribution is imbalanced.

- 'Matthews Correlation Coefficient' ('MCC') is one of the balanced performance measures, even for unbalanced datasets. It produces a correlation coefficient value that takes into account all four entries of the confusion matrix, T_P, T_N, F_P, and F_N. The output ranges between -1 and 1, where 1 denotes perfect classification, -1 denotes perfect misclassification, and a value around 0 denotes that the prediction is random.

The method of using these metrics to evaluate a machine learning model based on the values of the 'confusion matrix' for both binary and multi-class classification problems is given below [41, 42].

For binary classification

The calculation of the above measures for binary classification problems is presented in Table 3.

Table 3: Binary Classification

Metric	Formula
Accuracy	$(T\_N + T\_P)/(T\_N + F\_P + F\_N + T\_P)$
Precision	$(T\_P)/(F\_P + T\_P)$
Recall	$(T\_P)/(T\_P + F\_N)$
F1-Score	2 * (Recall * Precision) / (Recall + Precision)
MCC	$\frac{T\_P * T\_N - F\_P * F\_N}{\sqrt{(T\_P + F\_P)(T\_P + F\_N)(T\_N + F\_P)(T\_N + F\_N)}}$

For multi-class classification

All the above measures will be calculated separately for each class in a One-versus-Rest manner for multi-class classification problems, as shown in Table 4.

Table 4: Multi-class Classification

Metric	Formula
Accuracy	$\frac{\sum_{i=1}^{k} \frac{T\_P_i + T\_N_i}{T\_P_i + T\_N_i + F\_P_i + T\_N_i}}{k}$
Precision$_\mu$	$\frac{\sum_{i=1}^{k} T\_P_i}{\sum_{i=1}^{k} T\_P_i + F\_P_i}$
Recall$_\mu$	$\frac{\sum_{i=1}^{k} T\_P_i}{\sum_{i=1}^{k} T\_P_i + F\_N_i}$
F1-Score$_\mu$	$\frac{2 * Precision_\mu * Recall_\mu}{Precison_\mu + Recall_\mu}$
Precision$_M$	$\frac{\sum_{i=1}^{k} \frac{T\_P_i}{T\_P_i + F\_P_i}}{k}$
Recall$_M$	$\frac{\sum_{i=1}^{k} \frac{T\_P_i}{T\_P_i + F\_N_i}}{k}$
F1-Score$_M$	$\frac{2 * Precision_M * Recall_M}{Precison_M + Recall_M}$
MCC	$\frac{n_{correct} * n - \sum_{j=1}^{k} n_{j,predict} * n_{j,actual}}{\sqrt{(n^2 - \sum_{j=1}^{k} n_{j,predict}{}^2)} * \sqrt{(n^2 - \sum_{j=1}^{k} n_{j,actual}{}^2)}}$

In Table 4, the indices $_\mu$ and $_M$ denote Micro and Macro averages of measures, k is the number of target classes, n is the total number of samples, $n_{j,predict}$ is the number of samples predicted into class j, $n_{j,actual}$ is the actual number of samples in class j, and $n_{correct}$ is the total number of correctly predicted samples.

Micro ($_\mu$) and Macro ($_M$) averages represent two different ways to interpret confusion matrices in a multi-class problem. Macro average measures are calculated at the class level, giving equal importance to all classes. But micro average measures are calculated at the dataset level, giving equal importance to each unit rather than individual classes. So, macro averaging is the appropriate measure for a dataset with imbalanced class distributions.

3.6 Reviewing Model Outcome to Analyse the Effects of Emojis on Sentiment Classification

After evaluating the performances of various machine learning models, the next important step is to review the outcome of the evaluations to draw new inferences and conclusions. In this research work, the outcomes of evaluating ML models on three sets of input data (Text_alone, Emoji_alone and Text+Emoji) were reviewed. Since one of the research objectives was to demonstrate the effect of emojis and emoticons on sentiment classification, the performance of models on the 'Text_alone' set was compared with the other two sets, 'Emoji_alone' and 'Text+Emoji', and the results were analysed in Section 4.

3.7 Evaluation of ML Models on a Subset Causing Polarity Shifting to Analyse the Effects

Since the main objective of this research was to analyse the effects of emojis on polarity shifts and evolve an effective mechanism for sentiment classification, a subset of tweets showing such polarity shifts was identified and used to evaluate the same ML models in the next stage. The steps involved in this procedure are outlined below:

Step 1: Compute sentiment polarities for pure text (Text_alone).

Step 2: Consider the sentiment polarities for the original tweets with both Text and Emojis (Text+Emoji).

Step 3: Compare the polarities computed in Steps 1 and 2. (Comparisons with Observations are given in Table 5, and the impact of emojis on polarity shifting is categorised in the pie chart, as shown in Figure 3)

Step 4: Extract only the tweets in which emojis cause polarity shifting, reverse the polarity, or give meaning for pure neutral text (8% + 17% of cases from Figure 3).

Step 5: Train various ML classifiers listed in Section 3.4 on this subset of tweets (from Step 4) for all three sets ('Text_alone', 'Emoji_alone' and 'Text+Emoji'), and evaluate their performance (Accuracy is given in Table 8).

Table 5: Comparison of Sentiment Polarities of Pure Text with Original Text

Sentiment polarity of pure text	Sentiment polarity of original text with emojis	Observation	Percentage of cases
Positive Negative Neutral	Positive Negative Neutral	Emojis do not cause any change in the sentiment polarity	73%
Positive Negative	Negative Positive	Emojis reverse the sentiment polarity	8%
Neutral Neutral	Positive Negative	Emojis give meaning for pure neutral text	17%
Positive Negative	Neutral Neutral	Emojis neutralize the text	2%

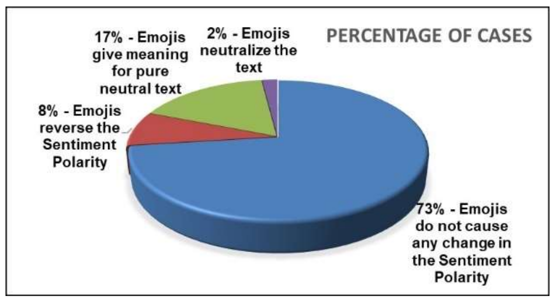


Figure 3: Effect of Emojis on the Input Tweets

3.8 Enhancing Emoji Sentiment Classification using EMOJI_SENT_WORD Method

To improve the performance of emoji sentiment classification, a new method named 'EMOJI_SENTI_WORD' has been proposed. The various stages of this method are illustrated in Figure 4.

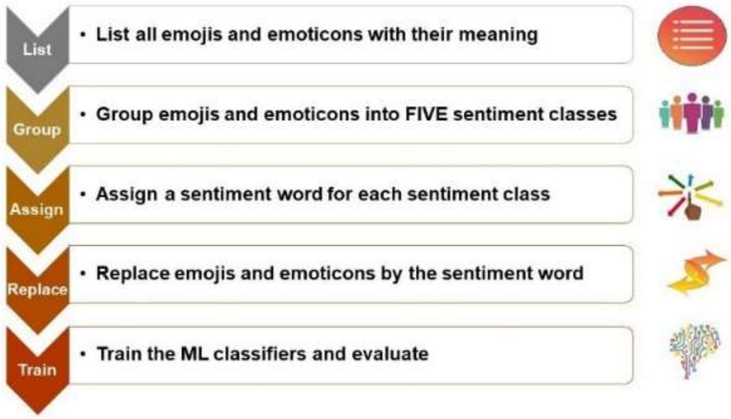


Figure 4: Outline of the Proposed Method 'EMOJI_SENTI_WORD'

In the traditional method, emojis and emoticons are converted from their Unicode into their corresponding CLDR (Common Locale Data Repository) short names [43] before training the ML classifiers. But in the proposed method 'EMOJI_SENTI_WORD', each occurrence of an emoji is replaced by a sentiment word chosen based on its intensity of sentiment polarity. The different stages of this proposed method are detailed below:

Stage 1: Various websites, such as emojipedia.org, emojiterra.com, and emoticonr.com, were referred to find a list of all available emojis and emoticons along with their meanings.

Stage 2: Emojis were grouped into FIVE sentiment classes based on their meanings: Highly Positive, Positive, Neutral, Negative, and Highly Negative.

Stage 3: Each sentiment class was assigned an appropriate sentiment word.

Stage 4: Input tweets were scanned for emojis, and when encountered, each emoji was replaced by the respective sentiment word based on its sentiment class.

Stage 5: After replacing all emojis with their respective sentiment words, the usual steps of training the ML classifiers and evaluating their performance were done.

Table 6 presents sample emojis from the input dataset grouped into five sentiment classes.

Table 6: Sample Emojis from Different Sentiment Classes

Sentiment class	Total number of emojis identified in the dataset	Sample emojis from the dataset
Highly Positive	24	
Positive	46	
Neutral	415	
Negative	30	
Highly Negative	6	

4. RESULTS AND DISCUSSION

This work has been implemented using Python libraries, namely Natural Language ToolKit (NLTK), sklearn, and emot. The research findings are presented in the following sub-sections, which analyse the effects of emojis on sentiment classification, especially when they cause polarity shifts, and demonstrate the efficiency of the proposed method in comparison to traditional methods. For improved clarity and comprehension, the results are represented through tables and charts in this section.

4.1 Effects of Emojis on Sentiment Classification (Traditional Vs Proposed Methods)

In Table 7, the performance of various machine learning classifiers evaluated on three sets (Text_alone, Emoji_alone and Text+Emoji) of original tweets, as mentioned in Section 3.4, using the 'GridSearchCV' hyperparameter optimization technique, is compared. The accuracy of these classifiers is then compared with the performance

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of the proposed method, ‘EMOJI_SENTI_WORD’, specifically for the Text+Emoji set.

Table 7: Comparison of Accuracies in both Traditional and Proposed Methods

ML classifier	Accuracy for Text_alone (%)	Accuracy for Emoji_alone (%)	Accuracy for Text+Emoji (%)	Accuracy of EMOJI_SENTI_WORD for Text+Emoji (%)
Stochastic Gradient Descent	65	65	80	82
Extreme Gradient Boosting	66	65	78	81
Logistic Regression	64	64	78	79
Support Vector Machine	63	63	77	80
Random Forest	65	65	76	81
Decision Tree	62	65	72	76
Naive Bayes	61	64	72	71
K Nearest Neighbor	49	62	67	68
Gradient Boosting	55	59	61	69
LSTM	**67**	**68**	**84**	**85**

The inferences from Table 7 are summarised below:

- LSTM, a type of RNN, demonstrates superior performance in all three sets: Text_alone, Emoji_alone, and Text+Emoji, both with the traditional method and with the proposed method (for the Text+Emoji set).
- The accuracies are highest in the third set (Text+Emoji), where the ML models are evaluated with both Text and Emojis. Applying the proposed method ‘EMOJI_SENTI_WORD’ to the Text+Emoji set has led to a significant improvement in accuracy, as shown in the last column of Table 7.
- The model evaluation with pure text (Text_alone with no emojis) yields the lowest accuracy values, proving the importance of emojis and their proper classification in improving sentiment classification accuracy.

In addition to the accuracy comparison, a comparison of the F1-Score percentages for all the above sets is also provided as a chart in Figure 5, which clearly shows that the proposed method ‘EMOJI_SENTI_WORD’ achieves higher F1-Scores when compared to the other three sets that used the traditional method.

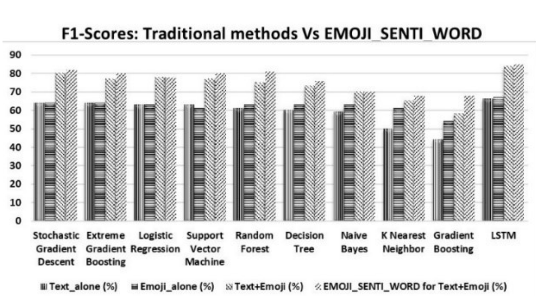

Figure 5: Comparison of F1-Scores in both Traditional and Proposed Methods

The percentage values of other performance measures, such as Precision and Recall, are also shown along with the F1-Score for the ‘EMOJI_SENT_WORD’ method in Figure 6:

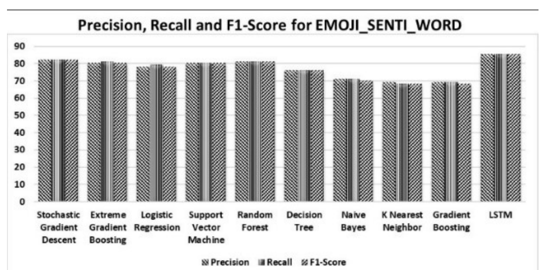

Figure 6: Precision, Recall and F1-Score of the Proposed Method

In one of the desirable performance measures for imbalanced datasets, the Matthews Correlation Coefficient (MCC), the values show a consistent increase from the Text_alone set to the Text+Emoji set, as depicted in Figure 7. Notably, the proposed method gives a much better result in this regard.

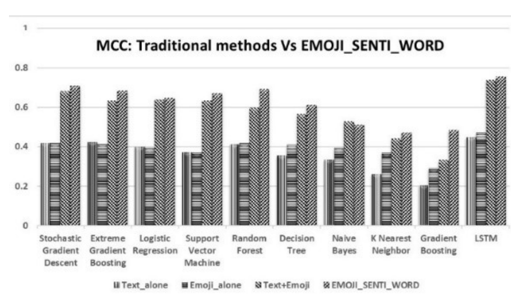

Figure 7: Comparison of MCCs in both Traditional and Proposed Methods

The Confusion Matrix for the 'EMOJI_SENTI_WORD' method applied to the Text+Emoji set, along with the predictions of all ML classifiers, is given in Figure 8. In this matrix, each row represents the 'actual class', while each column represents the 'predicted class', and hence, the diagonal entries of the matrix represent the 'correct predictions'.

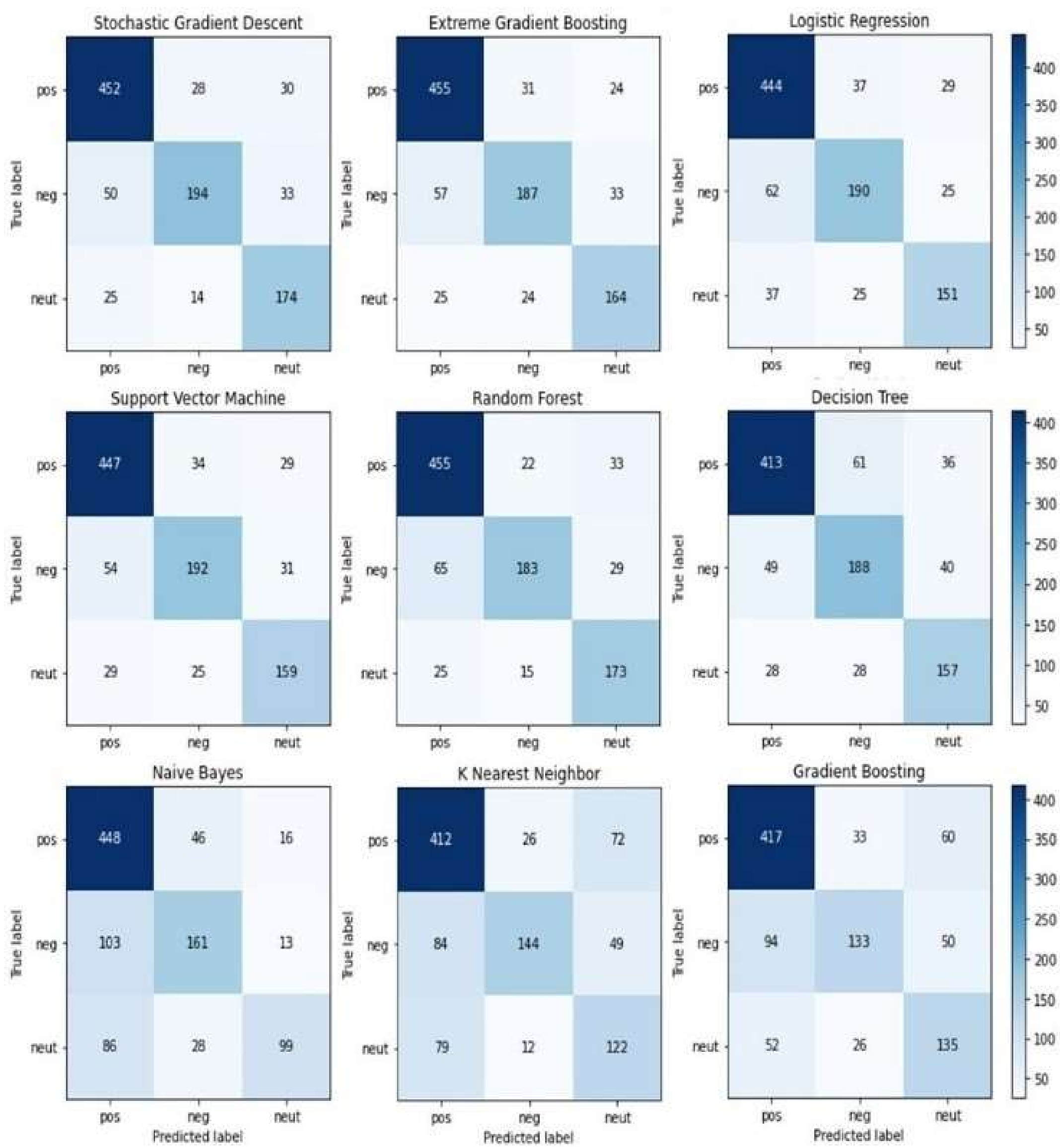

Figure 8: Confusion Matrix of the Proposed Method (for Text+Emoji)

4.2 Effects of Emojis on Polarity Shifting (Traditional Vs Proposed methods)

To demonstrate the effects of emojis on polarity shifts, a subset of tweets that caused polarity shifts alone was extracted (as explained in Section 3.7), and the same ML models were trained on three sets: Text_alone, Emoji_alone, and Text+Emoji, using the traditional method. In all cases, when compared to the results in Table 7 (of the original dataset), a significant increase in accuracy was observed across all classifiers, as presented in Table 8. Additionally, the proposed 'EMOJI_SENTI_WORD' method was applied on the Emoji_alone set, because this particular set exhibited relatively higher accuracy compared to the other two sets. The accuracies were found to be increasing up to 10% in the proposed method (for Gradient Boosting in Table 8).

Table 8: Comparison of Accuracies on the Subset of Tweets that Caused Polarity Shifting

ML classifier	Text_alone (%)	Text+Emoji (%)	Emoji_alone (%)	EMOJI_SENTI_WORD for Emoji_alone (%)
Stochastic Gradient Descent	73	94	94	97
Extreme Gradient Boosting	72	94	94	94
Logistic Regression	71	93	92	94
Support Vector Machine	75	92	94	94
Random Forest	73	94	93	94
Decision Tree	69	95	93	94
Naive Bayes	73	92	94	96
K Nearest Neighbor	71	90	93	94
Gradient Boosting	71	84	**84**	**94**
LSTM	75	94	**97**	**99**

Matthews Correlation Coefficient values have been plotted for the above sets in the chart given in Figure 9. In this chart, values closer to 1 denote perfect classification.

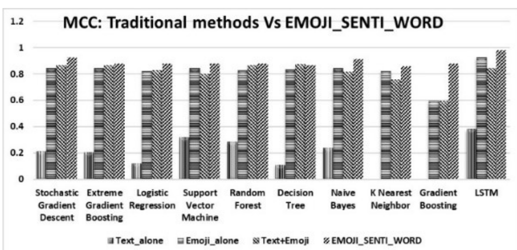

Figure 9: Comparison of MCCs on the Subset of Tweets that Caused Polarity Shifting

The important inferences of this work are summarised below:

- In the subset of tweets where emojis play a significant role in sentiment classification, they become the primary agent for inferring the real meaning of the text.
- Many texts where emojis cause polarity shifting are identified as being sarcastic, especially when there is a reversal of polarities.
- The highest accuracy can be achieved even when emojis alone are considered for sentiment classification, as illustrated in Table 8. It also reduces processing time compared to the original text classification, as explained in the following sub-section *'Effects on execution time'*.
- The effectiveness of incorporating emojis is not limited to a specific classifier, but applies to various classifiers. For instance, the 'Stochastic Gradient Descent classifier', which initially had an accuracy of 65% for pure text (Text_alone), was increased to 80% for Text+Emoji using the traditional method on the original dataset. However, when the proposed 'EMOJI_SENTI_WORD' method was applied to Text+Emoji, the accuracy increased to 82%. When trained with Emoji_alone for the subset of tweets causing polarity shifting, the accuracy achieved was 97%.

These findings highlight the importance of emojis in sentiment classification tasks. Ignoring or misclassifying emojis can significantly impact the accuracy and effectiveness of the system.

4.3 Effects on Execution Time

The execution time taken by one of the ML classifiers, Stochastic Gradient Descent, using the traditional method, is depicted in the chart (Figure 10). It presents the execution time in seconds plotted against different data sizes. It is clearly evident that the time taken with the Emoji_alone set is much less compared to the other two sets.

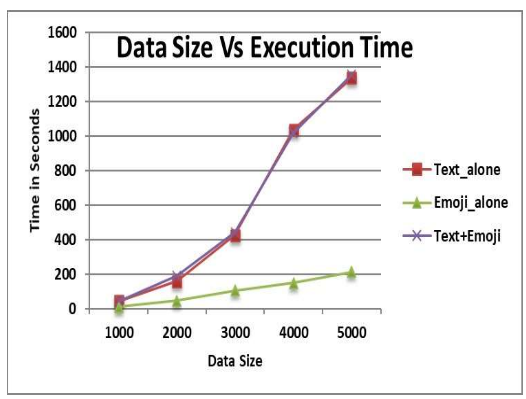

Figure 10: Execution Time of SGD for Different Data Sizes

A comprehensive overview of the execution times observed for all classifiers in the three sets (Text_alone, Emoji_alone, and Text+Emoji) of the original dataset using both traditional and proposed methods is also provided in Table 9.

ISSN: 1992-8645 www.jatit.org E-ISSN: 1817-3195

Table 9: Comparison of Execution Times of All Classifiers in Traditional and Proposed Methods

Classifiers	**Text_alone (in sec.)**	**Emoji_ alone (in sec.)**	**Text+Emoji (in sec.)**	**EMOJI_SENTI_ WORD Text+Emoji (in sec.)**	**EMOJI_SENTI_ WORD Emoji (in sec.)**
Stochastic Gradient Descent	1336.34	211.96	1353.28	15878.36	225.83
Extreme Gradient Boosting	41186.14	22468.87	43214.42	52800.27	2985.03
Logistic Regression	18.88	13.21	20.06	18.87	1.41
Support Vector Machine	508.23	67.75	529.44	2460.2	418.57
Random Forest	3785.95	1357.8	4662.02	6775.92	363.67
Decision Tree	98.14	10.87	52.54	23.89	0.91
Naive Bayes	8.08	7.66	8.49	2.21	1.08
K-Nearest Neighbor	13.16	7.79	12.34	6.16	4.62
Gradient Boosting	108.98	46.86	117.2	95.09	55.51
LSTM	542.38	552.44	456.39	548.95	70.38
TOTAL	**47606.28**	**24745.21**	**50426.18**	**78609.92**	**4127.01**

The total time taken by all classifiers in the five cases listed in Table 9 is visually represented in the chart (Figure 11).

Figure 11: Total Execution Time for the Three Sets in Traditional and Proposed Methods

The observations drawn from the execution time calculations for different sets are given below:

- Sentiment classification using emojis alone (Emoji_alone) has taken only approximately 50% of the execution time when compared to the Text_alone and Text+Emoji sets in the traditional method.
- In the proposed method 'EMOJI_SENTI_WORD', though the execution time for the Text+Emoji set is higher, it is found to be much lower for the Emoji_alone set.

These observations clearly demonstrate that employing emojis alone for sentiment classification greatly reduces the execution time compared to the conventional method of using both text and emojis.

4.4 Effects on Other Datasets

The results presented above were obtained by experimenting with a dataset of tweets collected using the filter 'corona'. To assess the effectiveness of the proposed method 'EMOJI_SENTI_WORD' in other domains, the same experiments were conducted on two additional datasets of tweets, extracted using the hashtags 'MovieReview' and 'OnlineClass'. The percentage of accuracy was found to be increasing up to 8% when using the proposed 'EMOJI_SENTI_WORD' method compared to the traditional method (Accuracy given in Table 10 for the Text+Emoji set).

Table 10: Accuracies in traditional and proposed methods for OnlineClass and MovieReview datasets

	OnlineClass Dataset		MovieReview Dataset	
ML classifier	**Accuracy for Text+Emoji (%)**	**Accuracy of EMOJI_SENTI_WORD for Text+Emoji (%)**	**Accuracy for Text+Emoji (%)**	**Accuracy of EMOJI_SENTI_WORD for Text+Emoji (%)**
Stochastic Gradient Descent	92	96	67	69
Extreme Gradient Boosting	87	88	**67**	**75**
Logistic Regression	**88**	**96**	72	75
Support Vector Machine	88	95	75	75
Random Forest	91	91	75	75
Decision Tree	88	89	64	69
Naive Bayes	80	83	72	75
K Nearest Neighbor	78	82	72	78
Gradient Boosting	45	45	67	67

Hence, the method proposed in this research for effective emoji classification could be applied across various fields of sentiment analysis, including social media analysis, psychology, health care, politics, education, sports, business, marketing, finance, stock market, and research, where emojis play a vital role in expressing feelings, emotions, and opinions.

4.5 Comparison with Earlier Research

The approximate accuracy of earlier works related to this research ranges from 42% to 79%. Although the accuracy achieved in this research is 7% to 8% higher, a direct comparison is not possible because the datasets used were not the same. Since none of those works use public benchmark datasets with emojis, this research compares the accuracy of the traditional method of emoji classification with the novel method proposed in this research, EMOJI_SENTI_WORD. As discussed thoroughly in this section, EMOJI_SENTI_WORD enhances system performance by reducing execution time and providing better accuracy in all scenarios. Comparison of execution time is also a hallmark in this research work, particularly in the realm of real-time applications with definite time constraints.

Some notable features of the proposed approach as compared to earlier research works are highlighted below.

- Holistic approach which includes both emoticons and emojis
- Adaptability to new emoticons and emojis easily with the help of "Sentiment Word" based classification
- Largest emoji dataset considering all the latest emojis used in the popular social media platforms
- Domain agnostic (tested across corona, movie review, and online education related datasets) and language independent
- Effective handling of sentiment classification with polarity shifting, with almost 8% higher accuracy than the current works
- Better performance compared to the other research works when Emoji alone is used, while the accuracy also improved from 97% to 99%. This provides the right balance between speed and accuracy

5. CONCLUSION AND FUTURE SCOPE

Sentiment analysis using emotions/ emojis plays a key role, especially in short tweets where polarity shift is influenced largely by the text/ graphical representation of emotions. Real-world applications like public opinion analysis, customer feedback analysis, targeted marketing, and trend analysis require a system that is scalable, adaptable, accurate, and performant. Current methods lack a holistic approach that can meet all the above requirements and play a crucial role in building Emotion AI systems. This research work addresses these challenges by adopting a structured approach to analyse various techniques and their shortcomings, and proposes a novel approach, 'EMOJI-SENTI_WORD' as explained below.

This research has thoroughly analysed the important role of emojis and emoticons in sentiment classification from various perspectives. The tweets extracted from Twitter were grouped into three sets: Text_alone (pure text excluding emojis), Emoji_alone (only emojis and emoticons), and Text+Emoji (original tweets with both text and emojis). When nine different machine learning classifiers and an LSTM deep learning network were trained on these sets, it was found that the accuracy of all classifiers was higher in the Text+Emoji set (with a maximum accuracy of 84%) compared to the Text_alone set (with 67% accuracy for the same classifier). This demonstrated the indispensable role of emojis in sentiment classification. The introduction of the 'EMOJI-SENTI_WORD' method to the Text+Emoji set further improved accuracy to 85%.

When the same classifiers were trained on the subset of tweets in which emojis caused polarity shifts, a maximum accuracy of 97% was achieved with the Emoji_alone set. With the implementation of the 'EMOJI_SENTI_WORD' method, accuracy was further increased to 99% for the same dataset. This indicates that emojis alone could be effectively used to identify the sentiment polarity of tweets with improved execution time. The execution time for the Text_alone set (47606.28 seconds) was reduced by half for the Emoji_alone set (24745.21 seconds). It decreased significantly (to 4127.01 seconds) when the EMOJI_SENTI_WORD method was applied to the Emoji_alone set. In most cases, the LSTM model resulted in high accuracy compared to other machine learning models.

The research has proved the effectiveness of the 'EMOJI_SENTI_WORD' method across different domains (including datasets filtered by 'corona', 'MovieReview', and 'OnlineClass'). This method can be used effectively to analyse sentiments in various fields where emojis are used to express emotions.

The performance of the proposed method can further be enhanced by increasing the number of sentiment classes beyond five. In the future, the context of tweets or texts can also be considered along with emojis to make more precise predictions. In the future, LLMs can also be used to automatically classify new emoticons/ emojis or to expand the classification categories for more accurate classification.

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